

24.3 Reliability and moment correction

This section supplies the missing information of [Section 24.2](#) directly, as a reliability ratio, an error covariance or a variance ratio, and solves Theorem 24.1.1 for θ .

24.3.1 Correcting with a known reliability

If λ is known, for instance from a test–retest correlation (Exercise (A1) in [Section 24.1](#)), then $\hat{\beta}/\lambda$ is consistent in simple regression. For correlations this is Spearman’s (1904) **correction for attenuation**, $\text{Corr}(X_1, X_2) = \text{Corr}(W_1, W_2)/\sqrt{\lambda_1\lambda_2}$ ([Exercise \(A1\)](#)). With covariates the right divisor is $\lambda_{x|z}$ ([Exercise \(B1\)](#)). Dividing by λ undercorrects and leaves the covariates’ bias untouched.

24.3.2 A known error covariance: the corrected estimator

Work with raw second moments, so the intercept needs no special treatment. Collect the true regressors, *including* the constant, in $\mathbf{t} = (1, \mathbf{x}^\top, \mathbf{z}^\top)^\top \in \mathbb{R}^p$, and the observed ones in $\mathbf{s} = \mathbf{t} + \tilde{\mathbf{u}}$, where $\tilde{\mathbf{u}} = (0, \mathbf{u}^\top, \mathbf{0}^\top)^\top$ has zeros for the constant and for the exactly measured $\mathbf{z}$. Write the model as $Y = \mathbf{t}^\top \theta + \varepsilon$. Under classical error,

$$E(\mathbf{s}\mathbf{s}^\top) = E(\mathbf{t}\mathbf{t}^\top) + \tilde{\Sigma}, \quad E(\mathbf{s}Y) = E(\mathbf{t}\mathbf{t}^\top)\theta, \quad \tilde{\Sigma} = E(\tilde{\mathbf{u}}\tilde{\mathbf{u}}^\top),$$

since the cross terms vanish. So $\theta = \{E(\mathbf{s}\mathbf{s}^\top) - \tilde{\Sigma}\}^{-1}E(\mathbf{s}Y)$: subtract the error’s share of the moment matrix before inverting. To cover a known $\tilde{\Sigma}$ and one estimated from replicates at once, let observation i carry a matrix $\mathbf{D}_i$, computed from its own data, with expectation $\tilde{\Sigma}$. If $\tilde{\Sigma}$ is known, $\mathbf{D}_i = \tilde{\Sigma}$.

Theorem 24.3.1 — Method-of-moments correction

Let $(\mathbf{t}_i, \tilde{\mathbf{u}}_i, \varepsilon_i, \mathbf{D}_i)$, $i = 1, 2, \dots$, be independent copies of $(\mathbf{t}, \tilde{\mathbf{u}}, \varepsilon, \mathbf{D})$, where $\mathbf{s}_i = \mathbf{t}_i + \tilde{\mathbf{u}}_i$ and $y_i = \mathbf{t}_i^\top \theta + \varepsilon_i$ are observed, and $\mathbf{D}_i$ is a symmetric $p \times p$ matrix computed from the data of observation i . Assume that $E(\varepsilon) = 0$, $E(\mathbf{t}\varepsilon) = \mathbf{0}$, $E(\tilde{\mathbf{u}}\varepsilon) = \mathbf{0}$, $E(\mathbf{t}\tilde{\mathbf{u}}^\top) = \mathbf{0}$, $E(\mathbf{D}) = \tilde{\Sigma} = E(\tilde{\mathbf{u}}\tilde{\mathbf{u}}^\top)$, and that $\mathbf{A} = E(\mathbf{t}\mathbf{t}^\top)$ is positive definite. Let

$$\hat{\mathbf{A}}_n = \frac{1}{n} \sum_{i=1}^n (\mathbf{s}_i \mathbf{s}_i^\top - \mathbf{D}_i), \quad \hat{\theta}_{\text{mm}} = \hat{\mathbf{A}}_n^{-1} \frac{1}{n} \sum_{i=1}^n \mathbf{s}_i y_i. \quad (24.3.1)$$

- (a) With probability one, $\hat{\mathbf{A}}_n$ is nonsingular for all large n and $\hat{\theta}_{\text{mm}} \rightarrow \theta$.
- (b) If $\mathbf{t}$, $\tilde{\mathbf{u}}$ and ε have finite fourth moments and $\mathbf{D}$ has finite second moments, then

$$\sqrt{n}(\hat{\theta}_{\text{mm}} - \theta) \rightarrow N_p(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}) \quad \text{in distribution,}$$

where $\mathbf{B} = E(\psi\psi^\top)$ and $\psi = \mathbf{s}(y - \mathbf{s}^\top \theta) + \mathbf{D}\theta$.

- (c) Under the conditions of (b), $\hat{\mathbf{A}}_n^{-1} \hat{\mathbf{B}}_n \hat{\mathbf{A}}_n^{-1} \rightarrow \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}$ in probability, where $\hat{\mathbf{B}}_n = n^{-1} \sum_i \hat{\boldsymbol{\psi}}_i \hat{\boldsymbol{\psi}}_i^\top$ and $\hat{\boldsymbol{\psi}}_i$ is $\boldsymbol{\psi}_i$ with $\hat{\boldsymbol{\theta}}_{\text{mm}}$ in place of $\boldsymbol{\theta}$.

Proof. (a) By the strong law of large numbers, $n^{-1} \sum_i \mathbf{s}_i \mathbf{s}_i^\top \rightarrow \mathbf{E}(\mathbf{s} \mathbf{s}^\top) = \mathbf{A} + \tilde{\boldsymbol{\Sigma}}$ (the cross terms vanish since $\mathbf{E}(\mathbf{t} \tilde{\mathbf{u}}^\top) = \mathbf{0}$) and $n^{-1} \sum_i \mathbf{D}_i \rightarrow \tilde{\boldsymbol{\Sigma}}$. So $\hat{\mathbf{A}}_n \rightarrow \mathbf{A}$, which is nonsingular, and $\hat{\mathbf{A}}_n$ is nonsingular for all large n because the determinant is continuous. Also $n^{-1} \sum_i \mathbf{s}_i y_i \rightarrow \mathbf{E}\{(\mathbf{t} + \tilde{\mathbf{u}})(\mathbf{t}^\top \boldsymbol{\theta} + \varepsilon)\} = \mathbf{A} \boldsymbol{\theta}$. By continuity of the inverse, $\hat{\boldsymbol{\theta}}_{\text{mm}} \rightarrow \mathbf{A}^{-1} \mathbf{A} \boldsymbol{\theta} = \boldsymbol{\theta}$.

(b) Directly from the definitions,

$$\frac{1}{n} \sum_i \mathbf{s}_i y_i - \hat{\mathbf{A}}_n \boldsymbol{\theta} = \frac{1}{n} \sum_i \{\mathbf{s}_i y_i - \mathbf{s}_i \mathbf{s}_i^\top \boldsymbol{\theta} + \mathbf{D}_i \boldsymbol{\theta}\} = \frac{1}{n} \sum_i \boldsymbol{\psi}_i,$$

so $\sqrt{n}(\hat{\boldsymbol{\theta}}_{\text{mm}} - \boldsymbol{\theta}) = \hat{\mathbf{A}}_n^{-1} n^{-1/2} \sum_i \boldsymbol{\psi}_i$. The $\boldsymbol{\psi}_i$ are independent and identically distributed with mean

$$\mathbf{E}(\boldsymbol{\psi}) = \mathbf{E}(\mathbf{s}Y) - \mathbf{E}(\mathbf{s} \mathbf{s}^\top) \boldsymbol{\theta} + \mathbf{E}(\mathbf{D}) \boldsymbol{\theta} = \mathbf{A} \boldsymbol{\theta} - (\mathbf{A} + \tilde{\boldsymbol{\Sigma}}) \boldsymbol{\theta} + \tilde{\boldsymbol{\Sigma}} \boldsymbol{\theta} = \mathbf{0}.$$

Each entry of $\boldsymbol{\psi}$ is a sum of products of at most two entries of $(\mathbf{t}, \tilde{\mathbf{u}}, \varepsilon)$, plus a linear function of $\mathbf{D}$, so the moment conditions give $\mathbf{B}$ finite entries. By the central limit theorem (the remark in [Section 14.4](#)), $\boldsymbol{\xi}_n = n^{-1/2} \sum_i \boldsymbol{\psi}_i \rightarrow N_p(\mathbf{0}, \mathbf{B})$. Split

$$\hat{\mathbf{A}}_n^{-1} \boldsymbol{\xi}_n = \mathbf{A}^{-1} \boldsymbol{\xi}_n + (\hat{\mathbf{A}}_n^{-1} - \mathbf{A}^{-1}) \boldsymbol{\xi}_n.$$

By continuous mapping ([Section 19.3](#)) the first term converges in distribution to $\mathbf{A}^{-1} N_p(\mathbf{0}, \mathbf{B}) = N_p(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$. Each entry of the second is a sum of products of an entry of $\hat{\mathbf{A}}_n^{-1} - \mathbf{A}^{-1}$, tending to 0, and an entry of $\boldsymbol{\xi}_n$; by the scalar Slutsky lemma it tends to 0 in probability. The sum form of Slutsky's lemma finishes the proof.

(c) Write $\mathbf{G}_i = \mathbf{s}_i \mathbf{s}_i^\top - \mathbf{D}_i$ and $\boldsymbol{\delta} = \hat{\boldsymbol{\theta}}_{\text{mm}} - \boldsymbol{\theta}$. Then $\hat{\boldsymbol{\psi}}_i = \boldsymbol{\psi}_i - \mathbf{G}_i \boldsymbol{\delta}$ and

$$\hat{\mathbf{B}}_n = \frac{1}{n} \sum_i \boldsymbol{\psi}_i \boldsymbol{\psi}_i^\top - \frac{1}{n} \sum_i (\boldsymbol{\psi}_i \boldsymbol{\delta}^\top \mathbf{G}_i^\top + \mathbf{G}_i \boldsymbol{\delta} \boldsymbol{\psi}_i^\top) + \frac{1}{n} \sum_i \mathbf{G}_i \boldsymbol{\delta} \boldsymbol{\delta}^\top \mathbf{G}_i^\top.$$

The first term tends to $\mathbf{B}$ by the strong law. Every entry of the second is at most $2\|\boldsymbol{\delta}\| n^{-1} \sum_i \|\boldsymbol{\psi}_i\| \|\mathbf{G}_i\|$ in absolute value (with the Frobenius norm for $\mathbf{G}_i$). The average converges to $\mathbf{E}(\|\boldsymbol{\psi}\| \|\mathbf{G}\|)$, which is finite by the Cauchy–Schwarz inequality, and $\|\boldsymbol{\delta}\| \rightarrow 0$ by (a). Similarly the third term is at most $\|\boldsymbol{\delta}\|^2 n^{-1} \sum_i \|\mathbf{G}_i\|^2 \rightarrow 0$. Combine with $\hat{\mathbf{A}}_n \rightarrow \mathbf{A}$.

■

In simple regression with known error variance, [Eq. 24.3.1](#) is $s_{wy}/(s_{ww} - \sigma_u^2)$, with sample moments taken with divisor n , the naive slope divided by the estimated reliability. The covariance $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}$ has the sandwich form of Theorem 21.4.2, to which it reduces when $\tilde{\mathbf{u}} = \mathbf{0}$ and $\mathbf{D} = \mathbf{0}$. No assumption about the distribution of X or homoscedasticity is needed.

Replicates. Suppose each unit has $m \geq 2$ readings $\mathbf{w}_{ij} = \mathbf{x}_i + \mathbf{u}_{ij}$, $j = 1, \dots, m$, with errors independent across j , independent of $(\mathbf{x}_i, \mathbf{z}_i, \varepsilon_i)$, and with covariance $\boldsymbol{\Sigma}_{uu}$. Use the mean $\bar{\mathbf{w}}_i$ as the observed regressor. Its error $\bar{\mathbf{u}}_i$ has covariance $\boldsymbol{\Sigma}_{uu}/m$, and an unbiased estimate of that

covariance from unit i alone is

$$\frac{1}{m(m-1)} \sum_{j=1}^m (\mathbf{w}_{ij} - \bar{\mathbf{w}}_i)(\mathbf{w}_{ij} - \bar{\mathbf{w}}_i)^\top,$$

padded with zeros to make $\mathbf{D}_i$ (Exercise A2)). For $m = 2$ and a scalar regressor this is $(w_{i1} - w_{i2})^2/4$. Because $\mathbf{D}_i$ enters $\boldsymbol{\psi}_i$, the sandwich of (c) includes the uncertainty from estimating the error variance. An external estimate treated as known would need that uncertainty added separately.

Example. (Correcting with two readings)

In the running example, regress y on $(1, \bar{w}, \text{age})$, where $\bar{w}$ is the mean of the two readings. The reliability of the mean is $144/(144 + 40.5) = 0.780$, and given age it is $108/(108 + 40.5) = 0.727$. The naive coefficients are 0.0562 (standard error 0.0054) for blood pressure and 0.0120 (standard error 0.0070) for age. The replicates estimate σ_u^2 as 74.1 (true 81), and Eq. 24.3.1 gives

$$\tilde{\beta}_x = 0.0757 \text{ (0.0076)}, \quad \tilde{\beta}_{\text{age}} = 0.0018 \text{ (0.0076)},$$

with sandwich standard errors. The 95% interval for β_x is [0.0608, 0.0905], and the spurious age effect is gone. The standard error is 1.40 times the naive one: the bias is paid for in variance. With $\sigma_u^2 = 81$ known the estimate would be 0.0782. In 10000 repeated samples the corrected intervals covered 0.08 in a proportion 0.949, the naive ones in 0.013.

```
import numpy as np

rng = np.random.default_rng(2401)
n = 300
age = rng.normal(55, 10, n)
x = 130 + 0.6 * (age - 55) + rng.normal(0, np.sqrt(108), n)  # long-run blood
                        pressure, sd 12
w = x[:, None] + rng.normal(0, 9, (n, 2))                  # two clinic readings
y = -4 + 0.08 * x + 0.0 * age + rng.normal(0, 1.0, n)      # age has no effect
                        given x

def corrected(w, age, y):
    """Moment-corrected coefficients of (1, x, age) and their sandwich standard
    errors."""
    n = len(y)
    W = np.column_stack([np.ones(n), w.mean(axis=1), age])  # observed regressors
    D = np.zeros((n, 3, 3))
    D[:, 1, 1] = (w[:, 0] - w[:, 1]) ** 2 / 4                # E D_i = Var(error
    of the mean)
    A = (W.T @ W - D.sum(axis=0)) / n
    theta = np.linalg.solve(A, W.T @ y / n)
    psi = W * (y - W @ theta)[:, None] + D @ theta         # estimating
    function, one row per patient
    B = psi.T @ psi / n
    Ainv = np.linalg.inv(A)
    cov = Ainv @ B @ Ainv / n
    return theta, np.sqrt(np.diag(cov))

theta, se = corrected(w, age, y)
```

```

W = np.column_stack([np.ones(n), w.mean(axis=1), age])
naive = np.linalg.lstsq(W, y, rcond=None)[0]
s2_u = np.mean((w[:, 0] - w[:, 1]) ** 2) / 2 # estimate of sigma_u
      ^2 (true 81)
print(f"sigma_u^2 estimate {s2_u:.1f}")
print("naive      (1, x, age):", naive.round(4))
print("corrected (1, x, age):", theta.round(4), " se:", se.round(4))

```

[Run this code online](#)

```

def coverage(reps, seed):
    """Coverage of nominal 95% intervals for beta_x = 0.08: corrected (sandwich) and
    naive."""
    r = np.random.default_rng(seed)
    hits_c = hits_n = 0
    for _ in range(reps):
        a = r.normal(55, 10, n)
        xx = 130 + 0.6 * (a - 55) + r.normal(0, np.sqrt(108), n)
        ww = xx[:, None] + r.normal(0, 9, (n, 2))
        yy = -4 + 0.08 * xx + r.normal(0, 1.0, n)
        th, s = corrected(ww, a, yy)
        hits_c += abs(th[1] - 0.08) <= 1.96 * s[1]
        Wr = np.column_stack([np.ones(n), ww.mean(axis=1), a])
        b = np.linalg.lstsq(Wr, yy, rcond=None)[0]
        e = yy - Wr @ b
        sb = np.sqrt(e @ e / (n - 3) * np.linalg.inv(Wr.T @ Wr)[1, 1])
        hits_n += abs(b[1] - 0.08) <= 1.96 * sb
    return hits_c / reps, hits_n / reps

print("coverage (corrected, naive):", coverage(500, 2421))

```

[Run this code online](#)

Warning. The theorem is asymptotic, and the correction is fragile when the reliability is low. $\hat{\mathbf{A}}_n$ can fail to be positive definite, and under normality the corrected slope has no finite mean ([Exercise \(C1\)](#)), as with Proposition 12.5.2; Fuller (1987) gives modified estimators. Gleser and Hwang (1987) showed that, as in Proposition 12.5.4, any confidence set with guaranteed coverage has infinite expected length. Wald intervals are reliable only when the estimated reliability is well away from zero.

24.3.3 Deming and orthogonal regression

Sometimes only a *ratio* of error variances is known. In a method-comparison study two instruments measure the same specimens, $W = X + U$ and $Y = \alpha + \beta X + \varepsilon$, with ε the second instrument's error, and repeatability experiments give

$$\delta = \frac{\text{Var}(\varepsilon)}{\text{Var}(U)}.$$

The question is whether $\alpha = 0$ and $\beta = 1$.

Proposition 24.3.2 — Deming regression

Let (W, Y) have covariances $\sigma_{ww}, \sigma_{wy}, \sigma_{yy}$ with $\sigma_{wy} \neq 0$ and $\sigma_{ww}\sigma_{yy} \geq \sigma_{wy}^2$, and let $\delta > 0$ be known.

- (a) The equations $\sigma_{ww} = \sigma_x^2 + \sigma_u^2$, $\sigma_{wy} = \beta\sigma_x^2$, $\sigma_{yy} = \beta^2\sigma_x^2 + \delta\sigma_u^2$ have exactly one solution with $\sigma_x^2 > 0$, namely

$$\beta_\delta = \frac{\sigma_{yy} - \delta\sigma_{ww} + \sqrt{(\sigma_{yy} - \delta\sigma_{ww})^2 + 4\delta\sigma_{wy}^2}}{2\sigma_{wy}}, \quad (24.3.2)$$

and for it $\sigma_u^2 \geq 0$. The slope β_δ lies in the interval of Proposition 24.2.2. It tends to b_2 as $\delta \rightarrow 0$ and to b_1 as $\delta \rightarrow \infty$.

- (b) For $\delta = 1$, β_1 is the slope of the line that minimizes the mean squared *perpendicular* distance from (W, Y) , whose direction is an eigenvector of the covariance matrix of (W, Y) for its largest eigenvalue. For general δ , β_δ is $\sqrt{\delta}$ times the $\delta = 1$ slope computed for $(W, Y/\sqrt{\delta})$.

Proof. (a) For $\beta \neq 0$ the first two equations give $\sigma_x^2 = \sigma_{wy}/\beta$ and $\sigma_u^2 = \sigma_{ww} - \sigma_{wy}/\beta$. Substituting in the third and multiplying by β ,

$$q(\beta) = \sigma_{wy}\beta^2 + (\delta\sigma_{ww} - \sigma_{yy})\beta - \delta\sigma_{wy} = 0.$$

The product of the roots is $-\delta < 0$, so there is one root of each sign, and $\sigma_x^2 > 0$ selects the root with the sign of σ_{wy} . That root is [Eq. 24.3.2](#). Take $\sigma_{wy} > 0$ (otherwise replace Y by $-Y$). Then

$$q(b_1) = \frac{\sigma_{wy}}{\sigma_{ww}^2}(\sigma_{wy}^2 - \sigma_{ww}\sigma_{yy}) \leq 0, \quad q(b_2) = \frac{\delta}{\sigma_{wy}}(\sigma_{ww}\sigma_{yy} - \sigma_{wy}^2) \geq 0,$$

and q is an upward parabola with $q(0) < 0$, so its positive root lies in $[b_1, b_2]$. Then $\sigma_u^2 = \sigma_{ww}(1 - b_1/\beta_\delta) \geq 0$. As $\delta \rightarrow 0$ the formula tends to $(\sigma_{yy} + \sigma_{yy})/(2\sigma_{wy}) = b_2$. Multiplying the numerator and denominator of [Eq. 24.3.2](#) by the conjugate of the numerator gives $\beta_\delta = 2\delta\sigma_{wy}/\{\sqrt{(\sigma_{yy} - \delta\sigma_{ww})^2 + 4\delta\sigma_{wy}^2} + \delta\sigma_{ww} - \sigma_{yy}\}$, and the denominator is $2\delta\sigma_{ww} + O(1)$, so $\beta_\delta \rightarrow b_1$ as $\delta \rightarrow \infty$.

(b) Let Σ be the covariance matrix of $\mathbf{v} = (W, Y)^\top$. A line through the point $E\mathbf{v}$ with unit normal vector $\mathbf{n}$ has mean squared perpendicular distance $E\{\mathbf{n}^\top(\mathbf{v} - E\mathbf{v})\}^2 = \mathbf{n}^\top \Sigma \mathbf{n}$, and a line through any other point $\mathbf{c}$ adds $\{\mathbf{n}^\top(E\mathbf{v} - \mathbf{c})\}^2 \geq 0$. By Theorem 1.7.7, $\mathbf{n}^\top \Sigma \mathbf{n}$ is minimized by an eigenvector for the smallest eigenvalue, so the line's direction is an eigenvector (v_1, v_2) for the largest eigenvalue μ . From $(\sigma_{ww} - \mu)v_1 + \sigma_{wy}v_2 = 0$, the slope is $v_2/v_1 = (\mu - \sigma_{ww})/\sigma_{wy}$. Put $b = v_2/v_1$. Then $\mu - \sigma_{ww} = b\sigma_{wy}$, and the second eigenvector equation $\sigma_{wy}v_1 + (\sigma_{yy} - \mu)v_2 = 0$ gives $\mu - \sigma_{yy} = \sigma_{wy}/b$. Subtracting, $\sigma_{yy} - \sigma_{ww} = b\sigma_{wy} - \sigma_{wy}/b$, which is $q(b) = 0$ with $\delta = 1$. Since $\mu \geq \sigma_{ww}$, b has the sign of σ_{wy} , so $b = \beta_1$. For the last claim, substitute σ_{yy}/δ and $\sigma_{wy}/\sqrt{\delta}$ for σ_{yy} and σ_{wy} in [Eq. 24.3.2](#) with $\delta = 1$ and multiply by $\sqrt{\delta}$. ■

With sample moments, [Eq. 24.3.2](#) is **Deming regression**, consistent because it is continuous where $\sigma_{wy} \neq 0$. With $\delta = 1$ it is **orthogonal regression** (total least squares), whose line is the

first principal axis of the scatter.

Warning. Orthogonal regression makes sense only when the errors have equal variances in the units used. Changing the units of Y changes the fitted line by more than a change of scale. Deming regression, with δ rescaled along with the units, does not have this problem.

Example. (Comparing two assays)

Sixty specimens with true concentrations uniform on $[2, 20]$ are measured by method A with error standard deviation 0.8 and by method B, which reads $0.3 + 1.05x$ with error standard deviation 1.0, so $\delta = 1.5625$. The four slopes are

Method	Slope
least squares of B on A	1.0042
Deming, $\delta = 1.5625$	1.0308
orthogonal ($\delta = 1$, unjustified)	1.0382
least squares of A on B, inverted	1.0709

with Deming intercept 0.482. Over 20000 simulated studies the mean slopes were 1.0262 for least squares, attenuated by $27/(27 + 0.64) = 0.9768$, and 1.0505 for Deming. With B in units ten times smaller, orthogonal regression gives 1.0703 (converted back to the original units) instead of 1.0382.

```
import numpy as np

def deming(w, y, delta):
    """Slope and intercept of Deming regression; delta = Var(error in y) / Var(error
    in w)."""
    S = np.cov(w, y)
    s_ww, s_wy, s_yy = S[0, 0], S[0, 1], S[1, 1]
    d = s_yy - delta * s_ww
    b = (d + np.sqrt(d ** 2 + 4 * delta * s_wy ** 2)) / (2 * s_wy)
    return b, y.mean() - b * w.mean()

rng = np.random.default_rng(2430)
x = rng.uniform(2, 20, 60)           # true concentrations
w = x + rng.normal(0, 0.8, 60)       # method A
y = 0.3 + 1.05 * x + rng.normal(0, 1.0, 60)  # method B
delta = 1.0 ** 2 / 0.8 ** 2

b_ls = np.polyfit(w, y, 1)[0]         # regress y on w
b_rev = 1 / np.polyfit(y, w, 1)[0]    # regress w on y, inverted
b_dem, a_dem = deming(w, y, delta)
b_orth, _ = deming(w, y, 1.0)         # orthogonal regression
print(f"least squares {b_ls:.4f}, Deming {b_dem:.4f} (intercept {a_dem:.3f}), "
      f"orthogonal {b_orth:.4f}, reverse {b_rev:.4f}")
```

[Run this code online](#)

24.3.4 Exercises

24.3.4.1 A. Check your understanding

Exercise A1.

Prove Spearman's correction: if $W_k = X_k + U_k$ for $k = 1, 2$, with classical errors independent of each other and of (X_1, X_2) , then $\text{Corr}(W_1, W_2) = \sqrt{\lambda_1 \lambda_2} \text{Corr}(X_1, X_2)$.

Exercise A2.

Show that the matrix $\mathbf{D}_i$ defined from m replicates has expectation $\mathbf{\Sigma}_{uu}/m$, the covariance matrix of $\bar{\mathbf{u}}_i$, padded with zeros.

Solution. $\mathbf{w}_{ij} - \bar{\mathbf{w}}_i = \mathbf{u}_{ij} - \bar{\mathbf{u}}_i$, so $\mathbf{x}_i$ drops out. For independent vectors with common covariance $\mathbf{\Sigma}_{uu}$, the sample covariance matrix $(m-1)^{-1} \sum_j (\mathbf{u}_{ij} - \bar{\mathbf{u}}_i)(\mathbf{u}_{ij} - \bar{\mathbf{u}}_i)^\top$ is unbiased for $\mathbf{\Sigma}_{uu}$ (Proposition 2.7.1). Dividing by m gives expectation $\mathbf{\Sigma}_{uu}/m = \text{Cov}(\bar{\mathbf{u}}_i)$.

24.3.4.2 B. Practice

Exercise B1.

For a scalar X and a scalar exactly measured Z , show that

$$\lambda_{x|z} = \frac{\lambda - \rho_{wz}^2}{1 - \rho_{wz}^2},$$

where ρ_{wz} is the correlation between W and Z . Check the formula on the running example, and deduce that $\lambda_{x|z} < \lambda$ whenever $\rho_{wz} \neq 0$ and $\lambda < 1$.

Solution. $\sigma_{x \cdot z}^2 = \sigma_x^2 - \sigma_{xz}^2/\sigma_z^2$, and since $\sigma_{wz} = \sigma_{xz}$, $\sigma_{xz}^2/\sigma_z^2 = \rho_{wz}^2 \sigma_w^2$. Divide the numerator and the denominator of $\lambda_{x|z} = \sigma_{x \cdot z}^2/(\sigma_{x \cdot z}^2 + \sigma_u^2)$ by σ_w^2 . The numerator becomes $\lambda - \rho_{wz}^2$ and the denominator $\lambda - \rho_{wz}^2 + (1 - \lambda) = 1 - \rho_{wz}^2$. In the example $\rho_{wz}^2 = 60^2/(225 \cdot 100) = 0.16$, so $\lambda_{x|z} = (0.64 - 0.16)/0.84 = 0.571$, as in [Section 24.1](#). Finally $(\lambda - \rho^2)/(1 - \rho^2) < \lambda$ iff $\lambda - \rho^2 < \lambda - \lambda \rho^2$ iff $\rho^2(1 - \lambda) > 0$.

Exercise B2.

In simple regression with centred variables and known σ_u^2 , take $\mathbf{D} = \text{diag}(0, \sigma_u^2)$ in [Theorem 24.3.1](#) and suppose (X, U, ε) are jointly normal and independent. Show that the asymptotic variance of $\sqrt{n}(\hat{\beta} - \beta)$ is

$$\frac{\sigma_w^2 \sigma_v^2 + \beta^2 \sigma_u^4}{\sigma_x^4}, \quad \sigma_v^2 = \sigma^2 + \beta^2 \sigma_u^2.$$

Compare it with σ^2/σ_x^2 , the asymptotic variance when X is observed. What does this give for the running example with the mean of two readings?

Solution. The slope's estimating function (the intercept decouples for centred variables) is $\psi = W(Y - W\beta) + \sigma_u^2\beta = Wv + \sigma_u^2\beta$ with $v = \varepsilon - \beta U$, and the slope entry of $\mathbf{A}$ is σ_x^2 . Now (W, v) is bivariate normal with $\text{Cov}(W, v) = -\beta\sigma_u^2$, and for such a pair $E(W^2v^2) = \sigma_w^2\sigma_v^2 + 2\text{Cov}(W, v)^2$. So $\text{Var}(\psi) = E(W^2v^2) - \{E(Wv)\}^2 = \sigma_w^2\sigma_v^2 + \beta^2\sigma_u^4$, and the sandwich is $\text{Var}(\psi)/\sigma_x^4$. It exceeds σ^2/σ_x^2 because $\sigma_w^2\sigma_v^2 > \sigma_x^2\sigma^2$. For the example, use $\sigma_u^2 = 40.5$ for the mean of two readings: $\sigma_w^2 = 184.5$, $\sigma_v^2 = 1 + 0.0064 \cdot 40.5 = 1.2592$, and the variance is $(184.5 \cdot 1.2592 + 0.0064 \cdot 1640.25)/144^2 \approx 0.01171$, a standard error of about $\sqrt{0.01171/300} \approx 0.0062$, below the 0.0076 of [Example \(Correcting with two readings\)](#), where σ_u^2 is estimated and age lowers the reliability.

24.3.4.3 C. Going deeper

Exercise C1.

In the normal structural model with known σ_u^2 and $n \geq 3$, show that $\tilde{\beta} = s_{wy}/(s_{ww} - \sigma_u^2)$ has $E|\tilde{\beta}| = \infty$. (Condition on $W_1, \dots, W_n$, and use the fact that s_{ww} has a continuous density that is positive at σ_u^2 .)

Solution. Given the W_i , (W_i, Y_i) is jointly normal and $E(Y | W) = a + \lambda\beta W$ with constant conditional variance $\tau^2 > 0$ (Theorem 3.5.2). So, given the W_i , $s_{wy} \sim N(\lambda\beta s_{ww}, \tau^2 s_{ww}/(n-1))$, and $E(|s_{wy}| | W) \geq \tau \sqrt{2s_{ww}/\{\pi(n-1)\}}$, because a normal variable's absolute mean is at least that of its centred version. Hence $E|\tilde{\beta}| \geq c E\{\sqrt{s_{ww}}/|s_{ww} - \sigma_u^2|\}$. Here $(n-1)s_{ww}/\sigma_w^2 \sim \chi^2(n-1)$, whose density is continuous and positive on $(0, \infty)$. So the density of s_{ww} is bounded below near σ_u^2 , and the integral of $1/|s - \sigma_u^2|$ diverges there.